

Utility Analysis Dashboard

Technical Design Note

Generated 2026-03-10 · Internal reference

1. Stack

Runtime: Python 3.11+, Streamlit, Plotly (go + px), pandas / numpy

Venv: shared ../finance_vis/venv_finance — do not create a new one

Launch:

```
source ../finance_vis/venv_finance/bin/activate && streamlit run app.py
```

Input: Korean .xslm/.xlsx files — buildings A/B/C/D, sheet 검침 내역

2. Module Map

```
app.py          <- Page config, top-level routing, file loading
sidebar.py      <- File loader, bins/tail stubs
data.py         <- Excel I/O (@st.cache_data), sheet constants, billing period
features.py     <- Column engineering: create_change_columns, aggregate_by_brand,
                  split_brand_by_floor, floor parsing, cols_brand_then_category
filters.py      <- render_meter_filters, show_filter_widgets, apply_sheet_filter
viz.py          <- plot_hist_with_tails (Plotly go.Bar histogram + IQR + bin slider)
utils_plot.py   <- bar_chart() shared Plotly bar helper (tab_cross + tab_efficiency)
lang.py         <- t() translation helper

meter_view.py   <- «검침내역» full pipeline: load -> filter -> histograms -> 3 tabs
summary.py      <- 요약 분석 view
billing.py       <- 수도광열비 부과 내역 view
ehp.py / water.py / hotwater.py / electricity.py <- per-sheet views
brand_profile.py <- Brand profile tab with peer comparison charts

tab_anomaly.py  <- 이상감지 tab (composite score, heatmap, spike/cost/HVAC)
tab_cross.py    <- 비용분석 tab (unit costs, electricity breakdown)
tab_efficiency.py <- 효율분석 tab (per-m2 usage benchmarking)
tab_corr.py / tab_reconciliation.py <- correlation & reconciliation tabs

anomaly_features.py <- build_anomaly_df: meter+billing+elec+water -> anomaly scores
cross_features.py <- build_unit_costs, build_elec_breakdown
biz_report.py   <- generate_anomaly/cross/efficiency_pdf + render_pdf_buttons()
report.py       <- generate_report_pdf (검침내역 main report)
billing_report.py / ehp_report.py / hvac_report.py <- per-sheet PDF generators
```

3. Navigation Architecture

```
sac.tabs (sidebar, left) -> nav_mode
+-- 시트 보기 -> sheet selectbox -> billing/ehp/water/hotwater/electricity/meter_view
+-- 분석      -> analysis selectbox
|   +-- 요약 분석 -> render_summary_view(water_df, hw_df, el_df)
|   +-- 업체 분석 -> render_meter_filters -> [이상감지|비용분석|효율분석] tabs
+-- 브랜드 프로필 -> render_brand_profile_tab
```

Key quirk: `sac.tabs` must render directly inside with `st.sidebar`: — never inside `st.empty()`. The empty placeholder causes the component to re-initialize on every rerun, losing state.

4. Data Pipeline (검침내역)

```
read_sheet()          # raw Excel, header=[2, 3, 4]
+- apply_header_rows() # flatten MultiIndex -> named columns
+- build_from_two_files() # prev file -> *_previous = prev month usage
+- create_change_columns() # *_change = curr-prev, *_pct
+- aggregate_by_brand() # group by (brand, building)
+- split_brand_by_floor() # divide by floor count
```

Cumulative vs usage: Meter readings are cumulative. `build_from_two_files` uses the pre-computed usage column (`water_usage_m3` etc.) as actual current-month consumption.

Column Naming

Column	Meaning
water_previous / water_current	This month's usage from prev/cur file
water_change	current - previous
water_pct	change / previous x 100
water_meter_prev/curr	Original cumulative readings (kept for reference)

5. Floor Logic

Floor values are compound strings: "1F/2F", "2F~5F", "B2F/B1F".

Function	Purpose
<code>parse_floor_value(s)</code>	Returns list of individual floor strings from compound value
<code>get_simple_floors(df)</code>	Sorted unique floors for multiselect widget
<code>split_brand_by_floor(df, sel_floors)</code>	Divides brand totals equally by floor count

When `sel_floors == ["All"]` and `sel_bldg == ["All"]` -> no splitting, use aggregate totals.

6. Histogram System (viz.py: `plot_hist_with_tails`)

Signature

```
plot_hist_with_tails(s, bins, lo, hi, title,
                    source_df=None, val_col=None, key="hist",
                    display_cols=None, tail_pct=None,
                    val_scale=1.0, show_bins_slider=True)
```

- `show_bins_slider=True` renders a Bins slider (5-200, default 50, step 5) above the chart. Pass `{mono('False')}` for callers that provide their own bins control (e.g. `{mono('_hist_controls')}` in `meter_view.py`).
- `val_scale` bridges when `s` is scaled (e.g. `/1e4` for 만원) but `source_df[val_col]` is raw 원:
`mask = (source_df[val_col] / val_scale >= x0) & (...<= x1)`

- Outlier table: separate top / bottom tails, columns: brand + val_col + {prefix}_current + {prefix}_previous + building + floor. Auto-derived when val_col ends in _change or _pct.
- Bin-click table: same column order as outlier table, shown directly below stats row.
- Style locked: blue normal bars (#4C72B0), amber tails (#DD8A00), red median line (#C44E52), white bg, height=380px, y-axis: Count (건)

IQR Controls (meter_view.py)

```
# Per-category tab:
bins, lo, hi = _hist_controls(key_prefix) # bins slider -> k slider -> equation
plot_hist_with_tails(..., bins=bins, show_bins_slider=False)

# All-category section:
_abins = st.slider("Bins", ...) # bins slider first
_ak = st.slider("IQR k", ...) # k slider second
st.markdown("$$Q_1=..., IQR=...$$ ...")
plot_hist_with_tails(..., bins=_abins, show_bins_slider=False)
```

7. Shared Bar Chart (utils_plot.bar_chart)

```
bar_chart(df, x, y, title, y_label,
          color_col="building", # str column name or None
          key=None, height=420)
```

Used by tab_cross.py and tab_efficiency.py — replaces duplicate _bar functions. Supports click-to-show-row interaction. Building colour map: A=#1f77b4, B=#d62728, C=#2ca02c, D=#9467bd.

8. PDF Generation Pattern

Shared helper: biz_report.render_pdf_buttons

```
render_pdf_buttons(
    pdf_key, # session_state key for cached bytes
    generator_fn, # lambda: generate_X_pdf(df)
    dl_label, # download button label
    dl_filename, # .pdf filename
)
```

All three biz tabs call this helper. PDF bytes are cached in session_state so regeneration is only triggered on button press, not on every rerun.

PDF button placement

Buttons appear at the top of each tab — immediately after data is loaded and KPIs are shown — so they are accessible without scrolling.

Raw data dropdown

Each tab shows a st.expander(" 원시 데이터") at the top with the underlying DataFrame — allows quick inspection of the data feeding the charts.

9. Money Formatting (원 units)

```
def _fmt_won(v):
    if abs(v) >= 1e8: return f"{v/1e8:,.0f} 억원"
    if abs(v) >= 1e4: return f"{v/1e4:,.0f} 만원"
    return f"{v:,.0f} 원"

_div, _unit = (1e8, "억원") if _max >= 1e8 else (1e4, "만원")
# pass val_scale=_div to plot_hist_with_tails so bin-click filter works
Rule: no decimals on 원 values — always :.0f.
```

10. Anomaly Scoring (anomaly_features.py)

Composite score = weighted sum of 5 components, each normalized to [0, 1]:

Component	Weight	Signal	Source
급등률	30%	SpikeMoM % change vs thresholds (100/50/20%)	검침내역
소비량	25%	Consolidant scores HH=4 HL=3 LH=2 Normal=1 LL=0	검침내역
비용	25%	Cost Z-scores of W/m3, W/kWh, 10kW/m2	수도광열비 부과 내역
HVAC	10%	kWh/m2 IQR-normalised	전체 전기 사용내역
일관성	10%	Consistency of zero-usage utilities	검침내역 +

Risk levels: 위험 >= 0.65 · 주의 >= 0.40 · 관찰 >= 0.20 · 정상 < 0.20

11. Common Pitfalls

#	Issue	Fix
1	Closure bug in loops	Default arg binding: def fn(key, _p=p, _cc=cc)
2	sac.tabs state loss	Never wrap in st.empty() — use with st.sid
3	Bin-click scale mismatch	Pass val_scale=factor to plot_hist_with_tails
4	Duplicate bins slider	Pass show_bins_slider=False when caller controls bins
5	Decimals on 원 (won)	Always :.0f, never :.2f
6	EHP column parsing	Cumulative cols M-DG (0-based 12-110); stop at next header
7	sac.tabs ValueError	Stored index >= tab count — clear nav_{file_name} key

12. Notable Session State Keys

Key	Purpose
nav_{file_name}	Sidebar nav tab index (sac.tabs)
anomaly_loaded_{file_name}	Gate: 이상감지 analysis started?
cross_loaded_{file_name}	Gate: cross-sheet data loaded?
anomaly_pdf_{file_name}	Cached anomaly PDF bytes

Key	Purpose
cross_pdf_{file_name}	Cached cross-tab PDF bytes
eff_pdf_{file_name}	Cached efficiency PDF bytes
{prefix}_iqr_k	IQR k multiplier per histogram section
{prefix}_bins / {prefix}_bins_i	Bins slider/input sync pair
all_bins_{p} / all_bins_i_{p}	Bins control for all-category histogram section

13. Pending / Known Issues

- 면적당 총비용 unit scaling: Display as 만원/m2 — reverted at user request, ready to re-implement.
- No other outstanding tasks as of 2026-03-10.